

SIGNALS AND SYSTEMS TEST I

Number of Questions: 35

Time: 90 min.

Directions for questions 1 to 35: Select the correct alternative from the given choices.

1. The step response of the system
 $y[n] = \beta y[n-1] + x[n]$, $-1 < \beta < 1$
 If the initial condition is $y[-1] = 1$

(A) $\frac{(1+\beta^{n+2})}{(1-\beta)}u[n]$

(B) $\frac{(1-\beta^{n+2})}{(1-\beta)}u[n]$

(C) $\left(\frac{1-\beta^{n+1}}{1-\beta}\right)u[n]$

(D) $\beta^{n+1}u[n] - \frac{1+\beta^{n+1}}{1-\beta}u[n]$

2. If a cosine signal is multiplied with a sinc signal then the resultant frequency characteristic look like

- (A) Band pass filter (B) Low pass filter
 (C) High pass filter (D) All pass filter

3. A random variable is known to have a cumulative distribution function $F_x(x) = u[x] \left[1 - \frac{x^2}{a}\right]$

Its density function is

(A) $u(x) - \frac{x}{a}\delta(x)$ (B) $u(x) \frac{2x}{a}e^{-x^2/a}$

(C) $u(x) \left[1 - \frac{x^2}{a}\right]\delta(x)$ (D) $\delta(x) - \left(\frac{2x}{a}\right)u(x)$

4. The impulse response of a discrete time system is given by $h[n] = \frac{1}{4}[\delta[n] + \delta[n-4]]$

The magnitude of the response can be expressed as

(A) $\frac{1}{4}|\cos 4\Omega|$ (B) $|\cos 4\Omega|$

(C) $\frac{1}{2}|\cos 2\Omega|$ (D) $|\cos 2\Omega|$

5. Match list - I & list - II

	List-I		List-II
1.	Odd signal	(P)	$x(n) = \left(\frac{1}{2}\right)^n u(n)$
2.	Energy signal	(Q)	$x(-n) = -x(n)$
3.	Periodic signal	(R)	$x(t)u(t)$
4.	Causal signal	(S)	$x(n) = x(n+N)$

- (A) $1-Q, 2-P, 3-S, 4-R$
 (B) $1-S, 2-R, 3-Q, 4-P$
 (C) $1-S, 2-R, 3-P, 4-Q$
 (D) $1-R, 2-P, 3-Q, 4-S$

6. The correct relation is

(A) $x(bt) \xrightarrow{F.T} bX(\omega/b)$

(B) $x(bt) \xrightarrow{F.T} bX(b\omega)$

(C) $x(t/b) \xrightarrow{F.T} bX(\omega/b)$

(D) $x(bt) \xrightarrow{F.T} 1/b X(\omega/b)$

7. Match list I & List II

	List-I		List-II
(P)	$y(n+4) + y(n+2) + y(n) = 4x(n+4) + x(n)$	1	Linear, time - invariant, dynamic
(Q)	$(n^2 + 1)y^2(n) + y(n) = 4x^2(n)$	2	Non - linear, time variable, dynamic
(R)	$y(n+3) + y(n+1) = 4x(n+1)$	3	Non - linear, time - variable, memory less
(S)	$y(n+3) + ny(n) = 3nx(n)$	4	Linear, time - variable, dynamic
		5	Non - linear, time - invariant, dynamic

Codes:

- (A) $P-2 \quad Q-3 \quad R-5 \quad S-1$
 (B) $P-2 \quad Q-5 \quad R-1 \quad S-4$
 (C) $P-3 \quad Q-5 \quad R-2 \quad S-1$
 (D) $P-3 \quad Q-2 \quad R-5 \quad S-4$

8. A linear system has the transfer function

$$H(j\omega) = \frac{1}{(j\omega + 2)}$$

When it is subjected to an input white noise process with a constant spectral density 'A' the spectral density of the output will be

(A) $\frac{A}{(j\omega + 2)}$ (B) $\frac{A}{\sqrt{\omega^2 + 4}}$

(C) $\frac{A}{(\omega^2 + 4)}$ (D) $\frac{A}{(\omega^2 + 2)}$

9. Consider two signals $x_1(t) = e^{20t}$ and $x_2(t) = e^{(-2+j)t}$
 Which one of the following statement is correct.

- (A) Both $x_1(t)$ and $x_2(t)$ are periodic.
 (B) $x_1(t)$ is periodic but $x_2(t)$ is not periodic
 (C) $x_2(t)$ is periodic but $x_1(t)$ is not periodic
 (D) Neither $x_1(t)$ nor $x_2(t)$ is periodic

10. Period of the sinusoidal signal

$x[n] = 10 \cos[0.5\pi n]$ is _____.

- (A) 3 (B) 4
 (C) 2 (D) 1

11. Which one of the following is a Dirichlet condition?

- (A) $\int_{t_1}^{\infty} |x(t)| < \infty$
 (B) signal $x(t)$ must have a finite number of Maxima and minima in the expansion interval.
 (C) $x(t)$ can have an infinite number of finite discontinuities in the expansion interval.
 (D) $x^2(t)$ must be absolutely summable.

12. A random sinusoidal signal $x(t) = \sin(\omega_0 t + \phi)$ where a random variable ' ϕ ' is uniformly distributed in the range $\pm \frac{\pi}{2}$. The mean value of $x(t)$ is

- (A) $\frac{2 \cos \omega_0 t}{\pi}$ (B) $\frac{2 \sin \omega_0 t}{\pi}$
 (C) zero (D) $\frac{2}{\pi}$

13. Consider the following statements regarding a linear discrete time system

$$H(z) = \frac{z^2 + 2}{(z+1)(z-0.2)}$$

- (1) The initial value $h(0)$ of the impulse response is -10 .
 (2) The system is stable
 (3) The steady state output is zero for a sinusoidal discrete time input of frequency equal to one – fourth the sampling frequency.

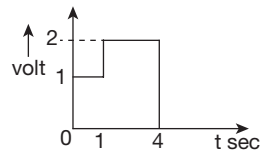
The correct statements are

- (A) 1, 2 and 3 (B) 1 and 2
 (C) 2 and 3 (D) 1 and 3

14. The discrete time system described by $y[n] = x(n^2)$ is

- (A) causal, linear and time varying.
 (B) non – Causal, linear and time – variant.
 (C) causal, non – linear, and time varying
 (D) non – causal, non – linear and time – variant.

- 15.



The Laplace transform of the waveform shown in the figure is _____

- (A) $\frac{1}{s} [1 + e^{-s} - 2e^{-4s}]$ (B) $\frac{1}{s} [1 + e^s - 2e^{4s}]$
 (C) $\frac{1}{s} [1 + e^{-s} - 2e^{+4s}]$ (D) $\frac{1}{s} [1 + e^{+s} - 2e^{-4s}]$

16. The sum of two or more arbitrary sinusoids is

- (A) Periodic under certain conditions
 (B) Never periodic
 (C) Always periodic

- (D) Periodic only if all the sinusoids are identical in frequency and phase.

17. The Nyquist rate for the signal $x(t) = \cos 1000\pi t + 4\sin 3000\pi t$

- (A) 2 KHz (B) 3 KHz
 (C) 6 KHz (D) 1 KHz

18. Magnitude & phase of the given network is



- (A) $\frac{1}{\omega^2}, 0$ (B) $\frac{1}{\omega^2}, \frac{\pi}{2}$
 (C) $\frac{1}{\omega^2}, \pi$ (D) $-\frac{1}{\omega^2}, 0$

19. If a LTI system s is given with impulse response $h[n]$ and Z – transform $H(z)$ and

- (1) $h[n]$ is real
 (2) $h[n]$ is left sided
 (3) $H(Z)$ has one of its poles at a non – real location on the circle defined by $|Z| = 2$

- (4) $H(Z)$ has two zeros

Then the system S is

- (A) non – causal & unstable
 (B) stable & causal
 (C) non – causal & stable
 (D) unstable & causal

20. If $x(t)$ be a signal with Nyquist rate ω_0 . Then the Nyquist rate for the signal $y(t) = x^2(t) \cos \omega_0 t + x^2(t) \sin \omega_0 t$ is

- (A) $8\omega_0$ (B) $2\omega_0$
 (C) $3\omega_0$ (D) $4\omega_0$

21. Two systems S_1 & S_2 are cascaded and their input – output relationship is given as

$$S_1 : y_1[n] = 3x_1[n-2] + 4x_1[n-4]$$

$$S_2 : y_2[n] = 5x_2[n-3] + 3x_2[n-5] + 7x_2[n-7]$$

If the overall system is S , then the overall impulse response $h[n]$ for the system S is —

- (A) $[0, 0, 0, 0, 0, 15, 20, 9, 21, 12, 0, 28]$

↑

- (B) $[0, 0, 0, 0, 0, 15, 0, 29, 0, 33, 0, 28]$

↑

- (C) $[15, 0, 29, 0, 33, 0, 28]$

↑

- (D) $[0, 0, 0, 0, 15, 0, 33, 0, 29, 0, 28]$

↑

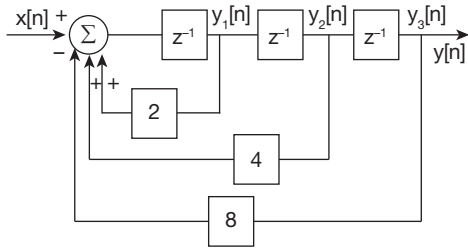
22. A signal $x(t)$ is given as

$$x(t) = \sin\left(\frac{3}{8}t + 30^\circ\right) + \cos\left(\frac{3}{4}t + 45^\circ\right)$$

Is periodic. The harmonics present in $x(t)$ are

- (A) Only 1st Harmonic
 (B) 1st and 2nd Harmonic
 (C) 2nd and 3rd Harmonics
 (D) 1st and 3rd Harmonics

Statement for Linked questions 23 and 24:

 An LTI system is given below for $n \geq 0$

 23. Transfer function $H(Z)$ for the system is

- (A) $\frac{z^{-3}}{2z^{-2} + 4z^{-3} + 8}$
 (B) $\frac{1}{z^3 - 2z^2 - 4z + 8}$
 (C) $\frac{1}{z^3 + 2z^2 + 4z - 8}$
 (D) $\frac{Z^{-3}}{1 - 2z^{-1} - 4z^{-2} - 8z^{-3}}$

 24. If $x[n] = \delta[n]$ then $y_3[n]$ is

- (A) $[1, 2, 8, 16]$ (B) $\left[\frac{1}{32}, \frac{1}{16}, \frac{1}{16}, \frac{1}{8}, \dots\right]$
 (C) $\left[\dots, \frac{1}{32}, \frac{1}{16}, \frac{1}{16}, \frac{1}{8}\right]$ (D) $[0, 0, 0, 1, 2, 8, 1, \dots]$

 25. A signal $x(t)$ has a duration of 5ms and an essential bandwidth of 25KHz and it is desirable to have frequency resolution of 125Hz in the DFT. The period N_o for discrete signal $x[n]$ is —

- (A) 100 (B) 200
 (C) 400 (D) 800

26. The phase spectrum of a function is

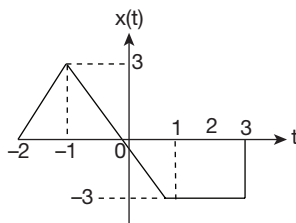
- (A) discrete function
 (B) odd function
 (C) symmetric function
 (D) anti-symmetric function

 27. $x[n]$ is a real periodic sequence, with period N and fourier co efficient of $x[n]$ is given as

 $C_K = a_k + jb_k$, if a_k, b_k are real then

- (A) $a_{-k} = -a_k, b_{-k} = b_k$ (B) $a_{-k} = a_k, b_{-k} = -b_k$
 (C) $a_{-k} = -a_k, b_{-k} = -b_k$ (D) $a_{-k} = a_k, b_{-k} = b_k$

28. The equation for the waveform shown in the figure is _____

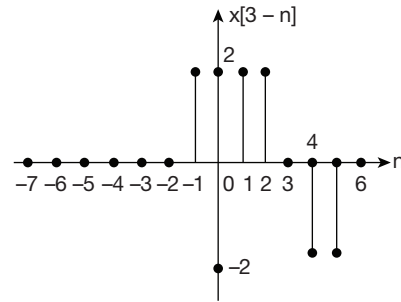
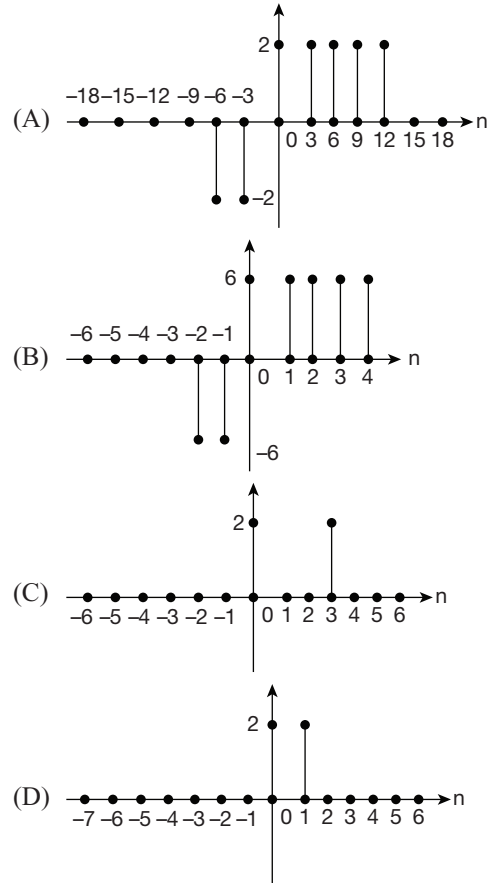


- (A) $3(t+2)u(t+2) - 6(t+1)u(t+1) + 3(t-1)u(t-1) + 3u(t-3)$
 (B) $3(t+2)u(t+2) - 3(t+1)u(t+1) + 2(t-1)u(t-1) + 3u(t-3)$
 (C) $3(t+2)u(t+2) + 3(t+1)u(t+1) + 3(t-1)u(t-1) - 2u(t-3)$
 (D) None of the above

 29. The impulse response of a continuous time LTI system is $h(t) = e^{-4t}u(4-t)$

The system is

- (A) stable but not causal
 (B) causal but not stable
 (C) causal and stable
 (D) neither causal nor stable

 30. The discrete time signal $x[3-n]$ shown below

 The signal $x[3n]$ is


31. Consider the three LTI systems with impulse response

$$h_1(t) = u(t)$$

$$h_2(t) = -2\delta(t) + 5e^{-2t}u(t)$$

$$h_3(t) = 2te^{-t}u(t)$$

The response to $x(t) = \cos t$ of above systems are

$$y_1(t) = x(t) * h_1(t)$$

$$y_2(t) = x(t) * h_2(t)$$

$$y_3(t) = x(t) * h_3(t)$$

which of the following gives same response

(A) $y_1(t)$ & $y_2(t)$

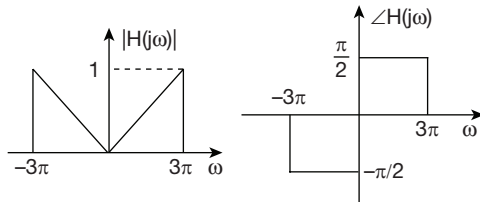
(B) $y_2(t)$ & $y_3(t)$

(C) $y_3(t)$ & $y_1(t)$

(D) All $y_1(t)$, $y_2(t)$ and $y_3(t)$

Common data for 32 and 33:

The frequency response $H(j\omega)$ of a continuous time filter is shown below.



32. If the input to this system is $x(t) = \cos(2\pi t + \theta)$, then output will be

(A) $\frac{-2}{3} \sin(2\pi t + \theta)$

(B) $\frac{2\pi}{3} \sin(2\pi t + \theta)$

(C) $\frac{2}{3\pi} \sin(2\pi t + \theta)$

(D) None of these

33. If the input to the system is $x(t) = \cos(4\pi t + \theta)$. Then the output will be

(A) $-\frac{4\pi}{3} \cos(4\pi t + \theta)$ (B) $-\frac{4}{3} \cos(4\pi t + \theta)$

(C) $\frac{4}{3\pi} \cos(4\pi t + \theta)$ (D) 0

34. Time delay of a sequence $x[n]$ is given by

$$D = \frac{\sum_{n=-\infty}^{+\infty} nx[n]}{\sum_{n=-\infty}^{+\infty} x[n]}$$

For $x[n] = \left(\frac{1}{3}\right)^n u[n]$, value of D at $\omega = 0$ is

(A) $\frac{2}{3}$ (B) 1

(C) 1/2 (D) 2

35. N -point DFT of $x[n] = u[n-N]$ is

(A) $X[k] = \begin{cases} 0 & \text{for } k \neq 0 \\ N & \text{for } k = 0 \end{cases}$

(B) $X[k] = 0$ for all k

(C) $X[k] = N$ for all k

(D) $X[k] = \begin{cases} N & \text{for } k \neq 0 \\ 0 & \text{for } k = 0 \end{cases}$

ANSWER KEYS

1. B	2. A	3. D	4. C	5. D	6. D	7. B	8. C	9. B	10. B
11. B	12. B	13. C	14. B	15. A	16. A	17. B	18. C	19. C	20. D
21. BB	22. B	23. D	24. D	25. C	26. C	27. B	28. A	29. D	30. D
31. D	32. A	33. D	34. C	35. A					

HINTS AND EXPLANATIONS

1. By taking z -transform on both sides,

$$Y(z) = \beta[(z^{-1} Y(z) + y(-1))] + X(z)$$

$$(1 - \beta z^{-1}) Y(z) = \beta + \frac{1}{1 - z^{-1}} \quad [\because x[n] \text{ is unit step}]$$

$$Y(z) = \frac{\beta}{1 - \beta z^{-1}} + \frac{1}{(1 - \beta z^{-1})(1 - z^{-1})}$$

By taking Inverse z -transform

$$y[n] = \beta \cdot \beta^n u[n] + \frac{(1 - \beta^{n+1})}{(1 - \beta)} u[n]$$

$$= \beta^{n+1} u[n] + \left(\frac{1 - \beta^{n+1}}{1 - \beta} \right) u[n] = \frac{(1 - \beta^{n+2})}{(1 - \beta)} u[n]$$

Choice (B)

2. Fourier transform of cosine signal is delta function.

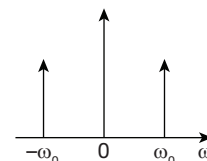
$\Rightarrow$ Fourier transform of sinc signal is gate function.

$\Rightarrow$ We know that multiplication in time domain is equal to convolution in frequency domain.

$\Rightarrow$ When delta function is convoluted with gate signal then the gate signal will itself present at the frequency where delta function was present.

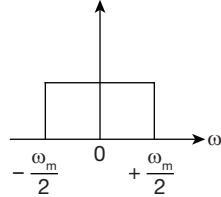
Let cosine signal is $\cos \omega_0 t$

F. T. of $\cos \omega_0 t$ is

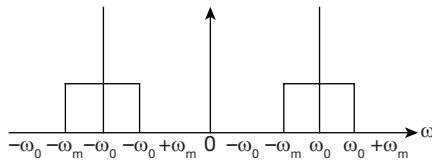


Let sinc signal is $\frac{\omega_m}{2\pi} \text{Sinc}\left(\frac{\omega_m t}{2}\right)$

Then F. T of $\frac{\omega_m}{2\pi} \left(\text{Sinc}\left(\frac{\omega_m t}{2}\right)\right)$ is



So after convolution
Resultant will look like



So like band pass filter.

Choice (A)

3. $CDF = F_x(x) = u[x] \left[1 - \frac{x^2}{a} \right]$

We know that $\frac{d}{dx}(CDF) = pdf$

$$\begin{aligned} \frac{d}{dx} F_x(x) &= \frac{d}{dx} \left[u[x] \left(1 - \frac{x^2}{a} \right) \right] \\ &= \left[\frac{d}{dx} u(x) \right] \left[1 - \frac{x^2}{a} \right] + u(x) \frac{d}{dx} \left(1 - \frac{x^2}{a} \right) \\ &= \delta(x) \left[1 - \frac{x^2}{a} \right] + u(x) \left(\frac{-2x}{a} \right) \cdot \left[\begin{array}{l} \delta(x) \cdot \frac{x^2}{a} = 0 \\ atx = 0 \end{array} \right] \\ &= \delta(x) - \frac{2x}{a} u(x) \end{aligned}$$

Choice (D)

4. $h[n] = \frac{1}{4} [\delta(n) + \delta(n-4)]$

Taking z - transform

$$H(z) = \frac{1}{4} [1 + z^{-4}]$$

Put $z = e^{j\Omega}$

$$H(e^{j\Omega}) = \frac{1}{4} [1 + e^{-j4\Omega}] = \frac{1}{4} e^{-j2\Omega} [e^{+j2\Omega} + e^{-j2\Omega}]$$

$$H(e^{j\Omega}) = \frac{1}{2} e^{-j2\Omega} [\cos 2\Omega]$$

$$H(e^{j\Omega}) = \frac{1}{2} |e^{-j2\Omega}| |\cos 2\Omega|$$

$$= \frac{1}{2} |\cos 2\Omega|$$

Choice (C)

- 5. • Odd signal $= x(n) = -x(-n)$
- Energy signal is absolutely summable

$$\sum_{n=0}^{\infty} \left| \left(\frac{1}{2} \right)^n u(n) \right| < \infty$$

- Periodic signal satisfies $x(n) = x(n + N)$
Where $N =$ Fundamental period
- Causal system is one which output at any time depends only on present and/ or past values of input.

Choice (A)

6. By scaling property

$$x(bt) \xleftrightarrow{FT} \frac{1}{b} X\left(\frac{\omega}{b}\right)$$

Choice (D)

7. Choice (B)

8. $S_0(\omega) = |H(\omega)|^2 Si(\omega)$

$$= \left[\frac{1}{(\omega^2 + 4)} \right] A = \frac{A}{(\omega^2 + 4)}$$

Choice (C)

9. $x_1(t)$ is periodic with period $\frac{\pi}{10}$ but $x_2(t)$ is not periodic
Choice (B)

10. $x[n] = 10\cos[0.5\pi n]$

$$\frac{2\pi}{N} = \omega$$

Where $N =$ time period

$$\text{So, } N = \frac{2\pi}{0.5\pi} = 4$$

Choice (B)

12. $\overline{x(t)} = \overline{\sin(\omega_0 t + \varphi)}$

$$= \int_{-\infty}^{+\infty} \sin(\omega_0 t + \varphi) \int \varphi(\varphi) d\varphi$$

$$= \frac{1}{\pi} \int_{-\pi/2}^{+\pi/2} \sin(\omega_0 t + \varphi) d\varphi$$

$$= \frac{1}{\pi} \left[-\cos(\omega_0 t + \varphi) \right]_{-\pi/2}^{+\pi/2}$$

$$= \frac{1}{\pi} \left[-\cos\left(\omega_0 t + \frac{\pi}{2}\right) + \cos\left(\omega_0 t - \frac{\pi}{2}\right) \right]$$

$$= \frac{1}{\pi} [\sin \omega_0 t + \sin \omega_0 t]$$

$$= \frac{2 \sin \omega_0 t}{\pi}$$

Choice (B)

13. (1) Roots of $H(z)$ are at $z = -1$ and $z = 0.2$ so the one root is inside the unit circle and second root is on the unit circle. So system is marginally stable.

- (2) Initial value of $h(t)$ by Initial value theorem

$$\lim_{t \rightarrow 0} h(t) = \lim_{Z \rightarrow \infty} H(z)$$

$$= \lim_{z \rightarrow \infty} \frac{z^2 + 2}{(z+1)(z-0.2)}$$

$$= \lim_{z \rightarrow \infty} \frac{1 + \frac{2}{z^2}}{\left(1 + \frac{1}{z}\right)\left(1 - \frac{0.2}{z}\right)} = 1 \quad \text{Choice (C)}$$

14. $y[n] = x[n^2]$

(i) For $n = -2$, $y(-2) = x(4)$

Since, $y[n]$ depends on future states also, it is a non-causal system

(ii) let $x_3[n] = ax_1[n] + bx_2[n]$
 $y_3[n] = x_3[n^2] = ax_1[n^2] + bx_2[n^2]$
 $= ay_1[n] + by_2[n]$

$\therefore$ system is linear

(iii) $x_1[n] \Rightarrow y_1[n] = x_1[n^2]$

Let $x_2[n] = x_1[n - n_0]$

$\Rightarrow y_2[n] = x_2[n^2] = x_1[n^2 - n_0]$

$y_1[n - n_0] = x_1[n - n_0]^2$

$y_2[n] \neq y_1[n - n_0]$

$\therefore$ The system is time variant.

Choice (B)

15. Let $v(t) = u(t) + u(t-1) - 2u(t-4)$

By taking laplace transform

$$V(s) = \frac{1}{s} + \frac{e^{-s}}{s} - \frac{2e^{-4s}}{s}$$

$$= \frac{1}{s} [1 + e^{-s} - 2e^{-4s}]$$

Choice (A)

16. Periodic under certain conditions.

Choice (A)

17. $x(t) = \cos 1000\pi t + 4\sin 3000\pi t$

Maximum frequency component

$$f_m = \frac{3000\pi}{2\pi} = 1500 \text{ Hz}$$

$$f_{\text{Nyquist}} = 2f_m$$

$$= 2 \times 1500 = 3000 \text{ Hz or } 3 \text{ KHz}$$

Choice (B)

18. $H(s) = H_1(s) \times H_2(s) = \frac{1}{s} \times \frac{1}{s}$

$$H(j\omega) = \frac{1}{j\omega} \times \frac{1}{j\omega} = \frac{1}{-\omega^2} = \left| \frac{1}{\omega^2} \right| e^{+j\pi}$$

$$\text{So } |H(j\omega)| = \frac{1}{\omega^2}$$

And $H(j\omega) = \pi$

Choice (C)

19. Since $h[n]$ is left sided sequence so system S is non-causal because ROC is inside the circle.

$\Rightarrow$ For stability, ROC should include unit circle & if it is left sided sequence & one of its poles at $|Z| = 2$ so ROC includes the unit circle so stable.

Choice (C)

20. $y(t) = x^2(t) \cos \omega_0 t + x^2(t) \sin \omega_0 t$

Let $x^2(t) = Z(t)$

Then $Z(t)$ will have double frequency in comparison to $x(t)$

Let maximum component present in $Z(t)$ is ω_z then $\omega_z = \omega_0 = 2\omega_x$

Where ω_x = maximum frequency component present in signal $x(t)$

$$Y(\omega) = \frac{1}{2} [Z(\omega_z + \omega_0) + Z(\omega_z - \omega_0)] + \frac{j}{2} [Z(\omega_z + \omega_0)$$

$$- (\omega_z - \omega_0)]$$

Maximum frequency component present is $Y(\omega)$ is $(\omega + \omega_0)$ and $\omega_z = \omega_0$

So ω_{max} in $Y(\omega) = 2\omega_0$

Nyquist rate = $4\omega_0$

Choice (D)

21. $y_1[n] = 3x_1[n-2] + 4x_1[n-4]$

$y_2[n] = 5x_2[n-3] + 3x_2[n-5] + 7x_2[n-7]$

$Y_1[z] = 3z^{-2} X_1(z) + 4z^{-4} X_1(z)$

$Y_2(z) = 5z^{-3} X_2(z) + 3z^{-5} X_2(z) + 7z^{-7} X_2(z)$

$H_1(z) = 3z^{-2} + 4z^{-4}$

$H_2(z) = 5z^{-3} + 3z^{-5} + 7z^{-7}$

$H_1(z)$ and $H_2(z)$ are cascaded

$H(z) = H_1(z) H_2(z)$

$$= (3z^{-2} + 4z^{-4}) (5z^{-3} + 3z^{-5} + 7z^{-7})$$

$$= 15z^{-5} + 9z^{-7} + 21z^{-9} + 20z^{-7} + 12z^{-9} + 28z^{-11}$$

$$H(z) = 15z^{-5} + 29z^{-7} + 33z^{-9} + 28z^{-11}$$

$h[n] = [\dots, 0, 0, 0, 0, 15, 0, 29, 0, 33, 0, 28, 0, 0, \dots]$

$h[n] = [0, 0, 0, 0, 0, 15, 0, 29, 0, 33, 0, 28]$

Choice (B)

22. $\therefore x(t)$ is periodic, so $x(t)$ is periodic with fundamental

period $\omega_0 = \text{LCM of } \left(\frac{3}{8}, \frac{3}{4}\right)$

Fundamental period $\omega_0 = \frac{3}{8}$

and $\omega_1 = \frac{3}{8}$ and $\omega_2 = \frac{3}{4}$

So Harmonic = $\frac{\omega_1}{\omega_0}$ and $\frac{\omega_2}{\omega_0}$

= 1st and 2nd Harmonics

Choice (B)

23. $y_3[n] = y_2[n-1]$ — (1)

$y_2[n] = y_1[n-1]$ — (2)

$y_1[n] = x[n-1] + 2y_1[n-1] + 4y_2[n-1] - 8y_3[n-1]$ — (3)

$y_3[n] = y_1[n-2]$ — (4)

= $x[n-3] + 2y_2[n-3] + 4y_2[n-3] - 8y_3[n-3]$

By eq (4) and (2)

$y_3[n] = x[n-3] + 2y_3[n-1] + 4y_3[n-2] - 8y_3[n-3]$

By taking z -Transform

$Y_2(z) = [1 - 2z^{-1} - 4z^{-2} + 8z^{-3}] = z^{-2} X(z)$

$$Y_3(z) = \frac{z^{-3}}{1 - 2z^{-1} - 4z^{-2} + 8z^{-3}}$$

$$= \frac{1}{z^3 - 2z^2 - 4z + 8}$$

Choice (B)

$$24. H(z) = \frac{1}{z^3 - 2z^2 - 4z + 8}$$

$$Y_3(z) = \frac{X(z)}{z^3 - 2z^2 - 4z + 8}$$

$$= \frac{1}{z^3 - 2z^2 - 4z + 8}$$

Since given LTI system is for $n \geq 0$ so it is causal system (Right sided sequence)

$$Z^3 - 2Z^2 - 4Z + 8$$

$$\frac{z^{-3} + 2z^{-4} + 8z^{-5} + 16z^{-6} \dots}{1}$$

$$\frac{1 - 2z^{-1} - 4z^{-2} + 8z^{-3}}{+ 2z^{-1} - 4z^{-2} + 8z^{-3}}$$

$$\frac{+ 2z^{-1} - 4z^{-2} + 8z^{-3} + 16z^{-4}}{8 - 2z^{-2} - 16z^{-4}}$$

$$\frac{8 - 2z^{-2} - 32z^{-4} - 16z^{-3} + 64z^{-5}}{+ 16z^{-3} + 16z^{-4} - 64z^{-5}}$$

$$Y_3(z) = Z^3 + 2Z^4 + 8Z^5 + 16Z^6 \dots$$

$$y_3[n] = [0, 0, 0, 1, 2, 8, 16, \dots] \quad \text{Choice (D)}$$

$$25. f_0 = 125\text{Hz}$$

$$\text{So } T_0 = \frac{1}{125} = 8\text{ms}$$

Since signal duration is of 8ms. So for 3ms, we use padding.

and Bandwidth $B = 25\text{KHz}$

So sampling frequency $f_s = 2B = 50\text{KHz}$

$$\text{So } N_o = \frac{f_s}{f_0} = \frac{50 \times 10^3}{125} = 400 \quad \text{Choice (C)}$$

26. Phase – spectrum of a function is symmetric function.

Choice (C)

27. Choice (B)

28. At $t = -2$, the slope of the signal changes from 0 to 3, for a change in slope 3.

At $t = -1$, the slope of the signal changes from 3 to -3 , for a change in slope of -6

At $t = 1$ the slope becomes 0 for a change of 3

At $t = 3$, the function steps from -3 to 0 for a change in amplitude of 3,

Hence equation for $x(t)$ is

$$3(t+2)u(t+2) - 6(t+1)u(t+1) + 3(t-1)u(t-1) + 3u(t-3) \quad \text{Choice (A)}$$

29. Not causal because $h(t) \neq 0$ for $t < 0$, unstable because

$$\int_{-\infty}^{+\infty} |h(t)| dt = \infty \quad \text{Choice (D)}$$

30. Let $x[3-n] = v[n]$

$$n = 0, x[3-0] = v[0] \Rightarrow x[3] = v[0] = 2$$

$$n = 1, x[3-1] = v[1] \Rightarrow x[2] = v[1] = 2$$

$$n = 2, x[3-2] = v[2] \Rightarrow x[1] = v[2] = 2$$

$$n = 3, x[3-3] = v[3] \Rightarrow x[0] = v[3] = 0$$

$$\text{Let } g[n] = x[3n] =$$

$$g[0] = x[0] = 0$$

$$g[1] = x[3] = 2$$

$$g[-1] = x[-3] = 0$$

Choice (D)

$$31. X(j\omega) = \pi [\delta(\omega+1) + \delta(\omega-1)]$$

$$H_1(j\omega) = \frac{1}{j\omega} + \pi\delta(\omega)$$

$$Y_1(j\omega) = \left[\frac{1}{j\omega} - \pi\delta(\omega) \right] \pi [\delta(\omega+1) + \delta(\omega-1)]$$

$$= j\pi [\delta(\omega+1) - \delta(\omega-1)] \quad (\text{at } \omega = 1 \text{ \& } \omega = -1 \text{ only})$$

$$y_1(t) = \sin(t)$$

$$H_2(j\omega) = -2 + \frac{5}{2+j\omega} = \frac{1-2j\omega}{2+j\omega}$$

$$Y_2(j\omega) = \left[\frac{1-2j\omega}{2+j\omega} \right] \pi [\delta(\omega+1) + \delta(\omega-1)]$$

$$= \left(\frac{1+2j}{2-j} \right) \pi\delta(\omega+1) + \left(\frac{1-2j}{2+j} \right) \pi\delta(\omega-1)$$

$$= j\pi \{ \delta(\omega+1) - \delta(\omega-1) \} \quad (\text{at } \omega = -1 \text{ \& } \omega = 1 \text{ only})$$

$$y_2(t) = \sin t$$

$$H_3(j\omega) = \frac{2}{(1+j\omega)^2}$$

$$Y_3(j\omega) = \frac{2}{(1+j\omega)^2} \pi \{ \delta(\omega+1) + \delta(\omega-1) \}$$

$$= \frac{2\pi}{(1-j)^2} \delta(\omega+1) + \frac{2\pi}{(1+j)^2} \delta(\omega-1)$$

$$= j\pi [\delta(\omega+1) - \delta(\omega-1)] \quad y_3(t) = \sin t$$

$$(\text{at } \omega = -1 \text{ \& } \omega = 1 \text{ only})$$

Choice (D)

$$32. H(j\omega) = \begin{cases} \frac{\omega}{3\pi} e^{\frac{j\pi}{2}}, & 0 \leq \omega \leq 3\pi \\ \frac{-\omega}{3\pi} e^{\frac{-j\pi}{2}}, & -3\pi \leq \omega \leq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow H(j\omega) = \begin{cases} \frac{j\omega}{3\pi}, & -3\pi \leq \omega \leq 3\pi \\ 0 & \text{otherwise} \end{cases}$$

$$x(t) = \cos(2\pi t + \theta)$$

$$X(j\omega) = e^{j\theta} \pi \delta(\omega-2\pi) + e^{-j\theta} \pi \delta(\omega+2\pi)$$

This is zero outside the region $-3\pi \leq \omega \leq 3\pi$

Thus $Y(\omega) = H(j\omega) \times (j\omega)$

$$= \frac{j\omega}{3\pi} X(j\omega)$$

$$y(t) = \frac{1}{3\pi} \frac{dx(t)}{dt} = \frac{-2}{3} \sin(2\pi t + \theta) \quad \text{Choice (A)}$$

33. $x(t) = \cos(4\pi t + \theta)$

$$X(j\omega) = e^{j\theta} \pi \delta(\omega - 4\pi) + e^{-j\theta} \pi \delta(\omega + 4\pi)$$

The non-zero positions of $X(j\omega)$ lie outside the range $-3\pi \leq \omega \leq 3\pi$. This implies that

$$Y(j\omega) = X(j\omega) H(j\omega) = 0, \text{ therefore } y(t) = 0 \text{ Choice (D)}$$

34. Let $X(e^{j\omega}) \xrightarrow{f} x[n]$

From the differentiation in frequency property of DTFT

$$\sum_{n=-\infty}^{+\infty} nx[n]e^{-j\omega n} = \frac{j dX(e^{j\omega})}{d\omega}$$

$$\text{So, } \sum_{n=-\infty}^{+\infty} nx[n] = \frac{j \cdot d \times (e^{j\omega})}{d\omega} \Big|_{\omega=0} \text{ ————— (1)}$$

DFTF of $x[n]$ is

$$x[e^{j\omega}] = \sum_{n=-\infty}^{+\infty} x[n]e^{-j\omega n}$$

$$\sum_{n=-\infty}^{+\infty} x[n] = X(e^{j0}) \text{ ————— (2)}$$

$$D = \frac{j \frac{dX(e^{j\omega})}{d\omega} \Big|_{\omega=0}}{X(e^{j0})}$$

$$\text{for } x[n] = \left(\frac{1}{3}\right)^n u[n]$$

$$X(e^{j\omega}) = F\left\{\left(\frac{1}{3}\right)^n u[n]\right\}$$

$$= \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

$$\text{So } \frac{j dX(e^{j\omega})}{d\omega} \Big|_{\omega=0} = \frac{\frac{1}{3}e^{-j\omega}}{\left(1 - \frac{1}{3}e^{-j\omega}\right)^2} \Big|_{\omega=0}$$

$$= \frac{\frac{1}{3}}{\left(1 - \frac{1}{3}\right)^2} = \frac{3}{4}$$

$$\text{and } X(e^{j0}) = \frac{1}{1 - \frac{1}{3}} = \frac{3}{2}$$

$$\text{So } D = \frac{\frac{3}{4}}{\frac{3}{2}} = \frac{1}{2}$$

Choice (C)

35. As we know that N-point DFT of $x[n]$ is

$$X[k] = \sum_{n=0}^{N-1} x[n] \omega_N^{kn}$$

$$= \sum_{n=0}^{N-1} \omega_N^{kn}$$

$$\frac{1 - \omega_N^{kn}}{1 - \omega_N^k} = 0, k \neq 0$$

$$\therefore \omega_N^{kn} = e^{-j(2\pi/N)kN} = e^{-jk2\pi} = 1$$

$$\text{And } X[0] = \sum_{n=0}^{N-1} \omega_N^0 = \sum_{n=0}^{N-1} 1 = N \text{ for } k=0 \text{ Choice (A)}$$